

THE BIRTH OF A UNIVERSE FROM A BLACK HOLE

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Date: 21.06.2026

Abstract

This episode opens Series 3 of VG, building on a central idea: **black holes are not an end, but a beginning**. When a black hole reaches critical coupling density, the Network does not collapse into an infinite singularity, but undergoes a **phase transition**—a local reconfiguration of spins and couplings, analogous to the unpacking of the Big Bang. From this transition, three possible destinies emerge: **pure absorption** (shell), **birth of a daughter Network** (bud), or **transformation into a white hole** (emission). This episode establishes the foundation for a new series exploring the **fractal cosmology of the Network**—a universe that reproduces itself through black holes at all scales.

Keywords: black hole, phase transition, daughter Network, imprinted rotation, fractal cosmology, VG Series 3.

1. INTRODUCTION – WHY A NEW SERIES?

- **VG Series 1** built the architecture of the Network.
- **VG Series 2** described the cosmic cycle—expansion, contraction, rebirth.
- **VG Series 3** will tell the story of reproduction—how the Network multiplies through black holes, giving birth to new universes.

This series is not an addition. It is a **natural extension**. If the Network can undergo a phase transition at a global scale (the Big Bang), then it can do the same locally—inside a black hole.

2. WHAT WE HAVE ESTABLISHED SO FAR

From VG Series 2:

- **BH** = a regime of extreme absorption. Couplings intensify, nodes draw closer.
- **Critical density** = a physical threshold, not an infinite singularity.
- **Quantum foam state** = maximum relational density, where space and time cease to exist as distinct entities.
- **Packed Network** = the state from which the Big Bang originated.

Now we add:

- **Daughter Network** = a new configuration born from the local phase transition.
- **Imprinted rotation** = inherited rotation from the parent Network.
- **Fractality** = the same process repeats at all scales.

3. THE PHASE TRANSITION MECHANISM – STEP BY STEP

- 1. Absorption** – Matter falls into the BH. Couplings K_{ij} intensify. Space curves, time slows.
- 2. Critical density** – A physical threshold is reached. The geometry can no longer be sustained.
- 3. Coupling reconfiguration** – Couplings reorganize. Nothing breaks—they rearrange. Nodes remain, but geometry dissolves.
- 4. Spin inversion and reorientation** – Aligned spins invert and reorient. Depending on local parameters (rotation, density), three scenarios can emerge:
 - Pure absorption – flux remains inward; the black hole slowly evaporates, leaving a shell.
 - Birth of a bud – spins unpack, triggering a mini-Big Bang that gives rise to a daughter Network.
 - Transformation into a white hole – spin inversion reverses the flux; the black hole becomes a source of emission.
- 5. Rotation imprint** – If the BH was rotating, the daughter Network or white hole inherits this rotation. It becomes a primordial torsion field.

4. FRACTAL IMAGE – NETWORKS WITHIN NETWORKS

This process is not unique. It repeats:

- At the scale of supermassive black holes.
- At the scale of stellar black holes.
- At the scale of quantum fluctuations (micro-black holes).

Thus, the universe is a **fractal**—a set of Networks within Networks, each born from a local phase transition. This is the cosmological vision of VG Series 3.

5. CONCLUSION – A NEW COSMOLOGY

VG Series 3 proposes a cosmology in which:

- **Black holes are not an end, but a beginning.**
- **The universe is fractal—Networks within Networks.**
- **Rotation and information are inherited from the parent Network.**
- **Dark matter is the remnant of previous cycles.**

This is our vision.

6. WHAT COMES NEXT

This episode opens Series 3. Future episodes will develop:

- **The mathematical formalism** – the equations of the phase transition.
- **Observational predictions** – signatures in the CMB and gravitational waves.
- **Connections to RTS** – how a reality slice is born from a black hole.

7. INVITATION

This framework is open. Any researcher interested in cosmology, quantum gravity, or the philosophy of science is invited to take it, develop it, and test it. Complete materials are available on open-access platforms under the authors' names.